Kristina P. Sinaga, PhD

Independent Researcher • PhD in Applied Mathematics
Machine Learning • Federated Learning • Multi-View Clustering

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RESEARCH PROFILE

Applied mathematician with a PhD from Chung Yuan Christian University (Taiwan) and postdoctoral experience at Chung Yuan Christian University and ISTI-CNR (Italy). Over seven years of experience in multi-view clustering, federated learning and privacy-preserving machine learning for heterogeneous data. Author of publications in IEEE, Elsevier and MDPI journals, with more than 3,870 citations (h-index 8). Seeking research positions in which mathematical methods for multi-source data are developed together with scientific applications.

RESEARCH INTERESTS

- **Clustering methodology:** k-means and fuzzy c-means algorithms for single-view and multi-view data, with learned view and feature weights and a focus on efficiency and accuracy for heterogeneous, high-dimensional data.
- **Pattern recognition and dimensionality reduction:** clustering-based feature selection and dimensionality reduction; scalable pattern recognition frameworks for high-dimensional data.
- **Federated learning and privacy-preserving AI:** adaptation of classical learning algorithms to multi-client, multi-view settings, with attention to privacy, data security and communication efficiency.

EDUCATION

PhD in Applied Mathematics
2016 – 2020
Chung Yuan Christian University, Taiwan

- **Dissertation:** Multi-View Fuzzy Clustering Algorithms for Multi-View Data (多視圖數據模糊聚類演算法)
- Graduated with honors. Published 2 journal articles (J1, J2) and 1 conference paper (C1) during doctoral studies and presented at 1 international conference. Further journal articles arising from the doctoral research were published after graduation (J3, J4, 2021).

MSc in Mathematics
2013 – 2015
University of Sumatera Utara, Indonesia

- **Thesis:** Stochastic Optimization Models for Emergency Service Location Problems (Model Optimasi Stokastik Penentuan Lokasi dan Jumlah Ambulan dengan Korelasi) (operations research, optimization theory, mathematical modeling).

BSc in Mathematics
2008 – 2013
University of Sumatera Utara, Indonesia

- **Thesis:** Analysis of the Influence of Gross Regional Domestic Product, Education, and Unemployment on Poverty in Regencies and Cities in North Sumatra(Analisis Pengaruh Produk Domestik Regional Bruto, Pendidikan dan Pengangguran terhadap Kemiskinan di Kabupaten/Kota Propinsi Sumatera Utara
- Graduated with high honors. Research project on regional economic analysis using mathematical modeling.

RESEARCH AND ACADEMIC APPOINTMENTS

Independent Researcher
2020 – Present
Self-directed research, conducted between and alongside the appointments below

- Developed new approaches to multi-view clustering and federated learning, and prepared multiple manuscripts for submission to journals.
- Obtained an arXiv endorsement and published preprints.
- Authored one journal article (J8, *Electronics*) and two preprints (P1, *arXiv* and P7, *Qeios*) on clustering and federated learning.

Postdoctoral Researcher October 2024 – September 2025
Institute of Information Science and Technologies (ISTI-CNR), Italy

- Designed and implemented new clustering algorithms for heterogeneous data, achieving 20–30% higher clustering accuracy on benchmark datasets.
- Authored five arXiv preprints (P2–P6) on clustering and other machine learning topics. Additionally, an article published in a scholarly journal that emerged from postdoctoral research conducted in Taiwan was subsequently released. The article was affiliated with ISTI CNR and CYCU (J7, *Electronics*).
- Authored two open-source software packages. These are PyPI packages named *mvkm-ed 1.1.0* and *gcomvkm 0.1.0*.

Postdoctoral Fellow March 2023 – March 2024
Department of Applied Mathematics, Chung Yuan Christian University, Taiwan

- Developed single-view and multi-view clustering algorithms in federated and non-federated settings, including multi-view k-means (MVKM) and multi-view fuzzy c-means (MVFCM); one of these works led to a publication in *IEEE TPAMI*.
- Advanced the theoretical foundations of privacy-preserving machine learning and federated clustering.
- Provided MATLAB implementations for multi-source and multi-client data, and validated the methods experimentally on public multi-view datasets.

Lecturer Specialist S3 November 2020 – March 2022
Bina Nusantara University, Indonesia

- Taught mathematics and data analysis courses to more than 180 undergraduate and graduate students, with a reported 35% increase in student engagement.
- Supervised more than 10 student research projects; developed and updated curriculum materials; introduced project-based learning and real-world case studies.

TEACHING AND ACADEMIC SERVICE

Master in Information Systems Management 2020 – 2022
Bina Nusantara University

- Business Intelligence and Analytics (regular and online programs): data-driven decision making, statistical analysis and business intelligence tools; 100+ graduate students.

Bachelor in Computer Science 2021 – 2022
Bina Nusantara University

- Calculus I (2021) and Discrete Mathematics (2021–2022); 80+ undergraduate students.

Moderator, International Guest Lecturer Series 2020 – 2021
Moderated 4 international academic events

- Coordinated guest presentations and facilitated discussions among students and faculty from universities in Indonesia and abroad.

Outreach and Mentoring 2020 – 2022
Various institutions

- STEM outreach for underserved communities and educational institutions; science communication events and public lectures on mathematical concepts.
- Mentoring of students across institutions; support for diversity and inclusion initiatives.

OPEN-SOURCE SOFTWARE

mvkm-ed 1.1.0 PyPI package
Multi-view clustering framework

- Combines Federated Multi-View K-Means (Fed-MVKM) with Rectified Gaussian Kernel Multi-View K-Means (MVKM-ED), a privacy-preserving distributed framework with enhanced discriminative power. Tested on synthetic data and the DHA (Depth-included Human Action) dataset.
- PyPI | GitHub

gcomvkm 0.1.0 PyPI package
Globally collaborative multi-view clustering

- Python implementation of Globally Collaborative Multi-View k-Means, which integrates collaborative transfer learning with entropy-regularized feature and view reduction. It dynamically eliminates uninformative components while balancing local view importance and global consensus. Tested on synthetic data.
- PyPI | GitHub

PEER REVIEW SERVICE

Reviewer since 2020 (40+ reviews) for the following journals:

- **IEEE:** *IEEE Access*, *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Knowledge and Data Engineering*.
- **Springer:** *Data Mining and Knowledge Discovery*, *Statistics and Computing*, *Journal of King Saud University – Computer and Information Sciences*, *The Journal of Supercomputing*.
- **ACM:** *ACM Transactions on Knowledge Discovery from Data*.
- **Elsevier:** *Computer Methods and Programs in Biomedicine*, *Information Fusion*, *Applied Soft Computing*.
- **Universal Wiser Publisher (UWP):** *Contemporary Mathematics*.
- **Intelligent Journal System (IJS):** *CMC: Computers, Materials & Continua*.
- **MDPI:** *Algorithms*, *Applied Sciences*, *Big Data and Cognitive Computing*, *Biomimetics*, *Designs*, *Diagnostics*, *Drones*, *Electronics*, *Entropy*, *Future Internet*, *Information*, *IoT*, *Journal of Imaging*, *Journal of Marine Science and Engineering*, *Machine Learning and Knowledge Extraction*, *Mathematics*, *Project Management Horizons*, *Sensors*, *Symmetry*.
- **Recognition:** MDPI *Mathematics* Exceptional Reviewers List, 2025.

AWARDS AND HONORS

- **2025** MDPI *Mathematics* Exceptional Reviewers List.
- **2020–present** Paper listed among the most-read documents in *IEEE Access*.
- **2020** Honorary Member, Phi Tau Phi Scholastic Honor Society, Chung Yuan Christian University, Taiwan.
- **2018** Japan Science and Technology Agency (JST) Scholarship, Niigata University, Japan.
- **2017** Japan Student Services Organization (JASSO) Scholarship, Niigata University, Japan.
- **2016** International Student Scholarship (full tuition and living expenses), Chung Yuan Christian University, Taiwan.

CERTIFICATIONS

Completed during a period of self-directed professional development (April 2022 – February 2023).

- **October 2022** Machine Learning Specialization, Coursera (Stanford University and DeepLearning.AI). Certificate
- **November 2022** TensorFlow Developer Specialization, Coursera (DeepLearning.AI). Certificate

TECHNICAL SKILLS

- **Programming and tools:** Python, R, MATLAB, Julia, C++, SQL, LaTeX, Git/GitHub, Docker, Linux/Unix.
- **Machine learning and scientific computing:** TensorFlow, PyTorch, scikit-learn, Keras, NumPy, Pandas, SciPy, Jupyter, Matplotlib, Seaborn.
- **Cloud and data infrastructure:** AWS, Google Cloud, Azure, Kubernetes, Apache Spark, Hadoop, MongoDB, PostgreSQL.

LANGUAGES

Bahasa Indonesia (native) • English (professional) • Mandarin (basic)

Last updated: October 2026

Publication List

Kristina P. Sinaga, PhD

Google Scholar | ORCID: 0009-0005-2230-8510 | GitHub

8 journal articles • 4 conference papers • 6 preprints • 1 doctoral dissertation • 1 master thesis • 1 undergraduate thesis • 3,870+ citations • 12 code repositories. Citation counts are approximate (Google Scholar). Own name in bold. Last updated: October 2026.

JOURNAL ARTICLES

J8 **Multi-View Clustering Goes Federated: A Survey**
K. P. Sinaga
Electronics (MDPI), 15(18):4103, 2026.
DOI

J7 **A Globally Collaborative Multi-View k-Means Clustering**
K. P. Sinaga and M. S. Yang
Electronics (MDPI), 14(11):2129, 2025.
DOI | Code | Citations: 10+

J6 **Federated Multi-View K-Means Clustering**
M. S. Yang and **K. P. Sinaga**
IEEE Transactions on Pattern Analysis and Machine Intelligence, 47(4):2446–2459, 2025.
DOI | Code | Citations: 50+

J5 **Unsupervised Multiview Fuzzy c-Means Clustering Algorithm**
I. Hussain, **K. P. Sinaga**, and M. S. Yang
Electronics (MDPI), 12(21):4467, 2023.
DOI | Code | Citations: 40+

J4 **Collaborative Feature-Weighted Multi-View Fuzzy c-Means Clustering**
M. S. Yang and **K. P. Sinaga**
Pattern Recognition, 119:108064, 2021.
DOI | Code | Citations: 95+

J3 **Entropy K-Means Clustering with Feature Reduction under Unknown Number of Clusters**
K. P. Sinaga, I. Hussain, and M. S. Yang
IEEE Access, 9:67736–67751, 2021.
DOI | Code | Citations: 60+

J2 **Unsupervised K-Means Clustering Algorithm**
K. P. Sinaga and M. S. Yang
IEEE Access, 8:80716–80727, 2020.
DOI | Code | Citations: 3,380+

J1 **A Feature-Reduction Multi-View k-Means Clustering Algorithm**
M. S. Yang and **K. P. Sinaga**
IEEE Access, 7:114472–114486, 2019.
DOI | Code | Citations: 160+

CONFERENCE PAPERS

C4 **Analytics-Based on Classification and Clustering Methods for Local Community Empowerment in Indonesia**
D. Yuniati and **K. P. Sinaga**
International Conference on Soft Computing in Data Science (ICSCDS 2021), pp. 133–145, 2021.
DOI

C3 **Telecommunication Analytics Based on Customer Segmentation Using Unsupervised Algorithms**
H. Wibowo and **K. P. Sinaga**
2021 3rd International Conference on Cybernetics and Intelligent System (ICORIS 2021), pp. 1–6, 2021.
DOI | Citations: 3

C2 **Machine Learning Approaches for Marketing Campaign in Portuguese Banks**
J. Alexandra and **K. P. Sinaga**
2021 3rd International Conference on Cybernetics and Intelligent System (ICORIS 2021), pp. 1–6, 2021.
DOI | Citations: 8

C1 **Modified Relational Mountain Clustering Method**
K. P. Sinaga, J. N. Hsieh, J. B. M. Benjamin, and M. S. Yang
Artificial Intelligence and Soft Computing: 17th International Conference (ICAISC 2018), pp. 690–701, 2018.
DOI | Citations: 3

PREPRINTS

P7 **Multi-View Clustering Goes Federated: A Survey**
K. P. Sinaga
Qeios: 10.32388/PC8X49.
DOI

P6 **Calibration Meets Reality: Making Machine Learning Predictions Trustworthy**
A. S. Nair and **K. P. Sinaga**
arXiv:2509.23665, 2025.
DOI | Code | Citations: 4

P5 **Personalized Federated Heat-Kernel Enhanced Multi-View Clustering via Advanced Tensor Decomposition Techniques**
K. P. Sinaga
arXiv:2509.16101 (v4), 2025.
DOI | Citations: 1

P4 **FedHK-MVFC: Federated Heat Kernel Multi-View Clustering**
K. P. Sinaga
arXiv:2509.15844 (v2), 2025.
DOI | Code | Citations: 5

P3 **Parameter-Free Entropy-Regularized Multi-View Clustering with Hierarchical Feature Selection**
K. P. Sinaga, S. Colantonio, and M. S. Yang
arXiv:2508.05504, 2025.
DOI | Code | Citations: 1

P2 **Bi-Level Multi-View Fuzzy Clustering with Exponential Distance**
K. P. Sinaga and S. Colantonio
arXiv:2503.22932, 2025.
DOI | Code | Citations: 2

P1 **Rectified Gaussian Kernel Multi-View k-Means Clustering**
K. P. Sinaga
arXiv:2405.05619, 2024.
DOI | Code | Citations: 5

DOCTORAL DISSERTATION

D1 **Multi-View Fuzzy Clustering Algorithms for Multi-View Data (**多視圖數據模糊聚類演算法**)**
K. P. Sinaga (許蒂娜)
PhD dissertation, Department of Applied Mathematics, Chung Yuan Christian University, Taiwan, pp. 1–92, 2020.
DOI

MASTER THESIS

M1 **Stochastic Optimization Models for Emergency Service Location Problems (**Model Optimasi Stokastik Penentuan Lokasi dan Jumlah Ambulan dengan Korelasi**)**
K. P. Sinaga
Master thesis, Faculty of Mathematics and Natural Sciences, Department of Mathematics, Universitas Sumatera Utara, Indonesia, 2015.
DOI

UNDERGRADUATE THESIS

U1 **Analysis of the Influence of Gross Regional Domestic Product, Education, and Unemployment on Poverty in Regencies and Cities in North Sumatra (**Analisis Pengaruh Produk Domestik Regional Bruto, Pendidikan dan Pengangguran terhadap Kemiskinan di Kabupaten/Kota Propinsi Sumatera Utara**)**

K. P. Sinaga

Undergraduate thesis, Faculty of Mathematics and Natural Sciences, Department of Mathematics, Universitas Sumatera Utara, Indonesia, pp. 1-68, 2013.

DOI